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Detecting potential outliers in longitudinal data with time-dependent covariates

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Abstract

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Author contribution statement

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Ethical approval

The TEDDY study was conducted in accordance with the Declaration of Helsinki, and approved by local US Institutional Review Boards and European Ethics Committee Boards, including the Colorado Multiple Institutional Review Board, Medical College of Georgia Human Assurance Committee (2004–2010), Georgia Health Sciences University Human Assurance Committee (2011–2012), Georgia Regents University Institutional Review Board (2013–2015), Augusta University Institutional Review Board (2015–present), University of Florida Health Center Institutional Review Board, Washington State Institutional Review Board (2004–2012), Western Institutional Review Board (2013–present), Ethics Committee of the Hospital District of Southwest Finland, Bayerischen Landesärztekammer (Bavarian Medical Association) Ethics Committee, Regional Ethics Board in Lund, Section 2 (2004–2012), and Lund University Committee for Continuing Ethical Review (2013–present).

Competing interests

The authors declare no conflict of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

Background: Outliers can influence regression model parameters and change the direction of the estimated effect, over-estimating or under-estimating the strength of the association between a response variable and an exposure of interest. Identifying visit-level outliers from longitudinal data with continuous time-dependent covariates is important when the distribution of such variable is highly skewed.

Objectives: The primary objective was to identify potential outliers at follow-up visits using interquartile range (IQR) statistic and assess their influence on estimated Cox regression parameters.

Methods: Study was motivated by a large TEDDY dietary longitudinal and time-to-event data with a continuous time varying vitamin B_{12} intake as the exposure of interest and development of Islet Autoimmunity (IA) as the response variable. An IQR algorithm was applied to the TEDDY dataset to detect potential outliers at each visit. To assess the impact of detected outliers, data was analyzed using the extended time-dependent Cox model with robust sandwich estimator. Partial residual diagnostic plots were examined for highly influential outliers.

Results: Extreme vitamin B_{12} observations that were cases of IA had a stronger influence on the Cox regression model than non-cases. Identified outliers changed the direction of hazard ratios, standard errors, or the strength of association with the risk of developing IA.

Conclusion: At the exploratory data analysis stage, the IQR algorithm can be used as a data quality control tool to identify potential outliers at visit-level which can be further investigated.

Keywords

Non-parametric statistics; Skewed data; Survival analysis; Nutrition; Extreme values

1. Introduction

In any statistical analysis, data is examined for unusual observations (1). Outliers can be defined as observations that are unusually larger or smaller as compared to other data points of the same variable (2). Several studies have suggested ways of defining outliers: they are observations that deviate extensively from the overall pattern or expectation of the other data points (see (3,4)), they are observations with large residual values (5) or they are data values falling outside of an expected range (6).

The cause of the extreme values may be the result of uncorrected data entry, system errors, self-reporting bias or they can be true observations that are due to rare events. Regardless, these outliers may have a large influence on the regression model parameters that can change the direction of the effect, mask the effect, underestimate the effect, or overestimate the effect (7,8).

Detection of outliers from longitudinal and time to event studies with continuous time-varying covariates provide a challenge in most applied research areas (9). Typically, the process of data cleaning and management takes a considerable amount of time due to complexity of large datasets. Yet, conducting exploratory data analysis to detect extreme values at an earlier stage of the statistical analysis is necessary to avoid misleading

conclusions that are based on only a few data points. There are several common approaches to detecting outliers including the use of knowledge and experience of investigators, using other published cross-sectional references or eyeballing using graphical outputs (10) to come up with single upper and/or lower values-cut-offs across the data set. Some studies (11) have used conditional growth percentiles to identify outliers in growth trajectory data and defined outliers to be observations 4 standard deviations (σ) away from the expected (conditional) value. Parameters, σ and mean (μ) are affected by extreme observations thus may not be suitable measures of spread and location, respectively, where the data is skewed (12). Additionally, conditional growth method cannot be applied to a subject's first measurement (visit). Robust regression methods have also been used to detect outliers on a longitudinal electronic health data records (13). Other methods include the use of Jackknife residuals (14) where they defined an outlier with a cut-off of ± 5 in their longitudinal childhood growth data, and studentized residuals (15) with a cut-off of $> 2\sigma$ to define outliers, whereas (16) used a cut-off of ± 6 for a longitudinal study on obesity prevalence and weight change in children and adolescents together with cross-sectional z-scores thresholds as defined by Centers for Disease Control.

Non-parametric statistics such as interquartile range (IQR) and median absolute deviation (MAD) are simple and alternative useful methods that can be applied to detect potential outliers during the exploratory data analysis stage. One study (17) calculated weight for length percentiles and then used the IQR method to detect and remove outliers in a longitudinal childhood genome study. The numerical IQR method followed by graphical assessment using box plots were described as an approach to successfully identify potential outliers in an educational achievement study to improve student learning (18). MAD method has been used to detect and remove outliers by several studies (see (12,19,20)). Both IQR and MAD statistics are robust measures of dispersion that are more resilient to outliers than standard deviation. The IQR method measures the central half of the data for any shape of the distribution and is a great alternative measure of dispersion that does not require symmetry and has no distribution assumptions since it uses percentiles making it more robust to the presence of outliers.

The primary aim of this study was to describe how to identify potential outliers for longitudinal data with skewed distributions at follow-up visits using a non-parametric interquartile range statistic method. This method may be a specific tool/procedure for the statistical analyst with no prior additional knowledge or policy at hand for dealing with extreme values.

2. Materials and Methods

2.1. Robustness of IQR and MAD statistics

The robustness of IQR and MAD has been shown by (21) that if T_n is a statistic on an ordered sample of size n , then, T_n has breakdown value b , $0 \leq b \leq 1$, if for every $\epsilon > 0$, $\lim_{X(\{(1-b)n\} \rightarrow \infty)} T_n < \infty$ and $\lim_{X(\{(1-(b+\epsilon)n\} \rightarrow \infty)} T_n = \infty$. The sample median (m) remains unchanged in the presence of extreme low or high values. If less than 50% of the sample $\rightarrow \infty$, then m and MAD will remain the same. If more than 50% of the sample $\rightarrow \infty$, then

$m \rightarrow \infty$ and so does MAD. The median in that case will be located within the outliers and thus MAD has a breakdown value of 50%. Similarly, IQR has a breakdown value of 25%, breaking down when Q_1 is located within the outliers (22).

2.2. Interquartile range algorithm

For each continuous time-dependent variable of interest, the following defines the IQR algorithm and can be applied overall or stratified by important subject level factors such as country or longitudinal variables such as age or follow-up visit:

- i. Calculate $IQR = Q_3 - Q_1$.
- ii. Define lower and upper limits of outliers as $[Q_1 - k \times IQR, Q_3 + k \times IQR]$ or $[Q_1 - h, Q_3 + h]$ where $h = k \times IQR$ and $k > 0$ is a scale factor.
- iii. Flag observations outside the limits as potential outliers.

IQR lower-limits below zero can be set to zero, for instance, in food record data where intake cannot be negative.

2.3. Motivating TEDDY dietary data

This study was motivated by a large longitudinal and time-to-event data set with time varying covariates. Dietary intake data was obtained from The Environmental Determinants of Diabetes in the Young (TEDDY), which is an observational longitudinal study that investigates factors associated with Diabetes (T1D) in children (23). Of the 8,676 children that were enrolled in the TEDDY study, 120 were HLA ineligible, 22 had no food records, and 33 had no record on Islet Autoimmunity and were dropped, leaving a total of 8,501 subjects with 152,426 records followed up from birth up until censoring at 10 years of age for this study. Diet was assessed by 24-hour dietary recall at the age of 3 months, by 3-day food record at the age of 6, 9, and 12 months and every 6 months thereafter until the subject developed islet autoimmunity (IA) or was censored at the end of the study period. For the purpose of illustrating the IQR method, focus was on exposure to daily intake of vitamin B_{12} ($\mu\text{g/day}$), including intake from foods and dietary supplements (24) and the risk of developing IA.

2.4. Statistical Analysis

The basic Cox proportional hazards model (25) (see also (26,27)) assumes that exposure covariates are fixed. This model can be extended to introduce variables which vary continuously with time in the form of $z_i(t) = z_i g(t)$ and expressed as

$$\begin{aligned} h(t) &= h_0(t) \exp \left\{ \beta_1 x_1 + \dots + \beta_k x_k + g(t) (\gamma_1 z_1 + \dots + \gamma_m z_m) \right\} \\ &= h_0(t) \exp \left(\sum_{i=1}^k \beta_i x_i + \sum_{i=1}^m \gamma_i z_i(t) \right). \end{aligned}$$

where $Z = \{z_1, \dots, z_m\}$ are time-varying covariates and γ_i are regression coefficient for a covariate $g(t)z_i$, which is a function of time. With the data arranged using the counting process, the extended time-dependent cox model associating the risk of islet auto-immunity and exposure variables is expressed as

$$h(t|X, Z) = h_0(t) \exp(\beta_1 \text{vitamin} B_{12}(t) + \beta_2 \text{sex} + \beta_3 \text{fdr} + \beta_4 \text{hla} + \beta_5 \text{country})$$

where vitamin B_{12} ($\mu\text{g/day}$) is the continuous time varying exposure of interest, adjusted for the fixed effects covariates: sex, HLA DR3/4, FDR, and country. Standard errors (SE) were calculated using robust (empirical) variance sandwich estimator to account for correlations within subjects (28,29). Vitamin B_{12} ($\mu\text{g/day}$) was energy adjusted by country and visit using Willett's residual method (30).

2.5. Detecting highly influential observations using residual diagnostics

Residual diagnostics plots were obtained after fitting Cox regression models on the full TEDDY dietary data to further ascertain the influence of highly influential observations on the estimated parameters. Partial DFBETA measure of influence for vitamin B_{12} was plotted against analysis time and partial efficient score residuals were also plotted against analysis time to identify observations with disproportionate influence. Partial residuals were calculated for each observation within the subject. These are the additive contributions to a subject's overall residual (see (26,27) for details). The partial DFBETA value estimates the change in the regressor's coefficient due to deletion of that individual record.

Although the primary aim of this study is to describe methods to identify outliers, we provide some guidelines on how to handle potential outliers. The steps below were followed in this study:

- i. The full data set was analyzed with the presence of these outliers for sensitivity analysis.
- ii. The variable was log-transformed to base 2 to avoid dropping observations.
- iii. Removed outliers based on IQR scale factors $k = 3, 5, 7, 10$.

Detected outliers for $k = 3$ were inversely weighted so that they can lie within the lower and upper bounds using the following procedure: For each x_i value that is outside the limits, compute weight $w_i = \max(|x_i - \text{upper limit}|, |x_i - \text{lower limit}|)$. Calculate a pseudo value $x_j = \frac{x_i}{w_i} + v$, where $v > 0$ is any constant (such as lower limit, $Q3$, etc.) such that x_i outlier is replaced with x_j that is bounded within the limits. An alternative is to replace x_i with the group-specific quartile value such as the median or a combination of quartiles ($Q1 + Q2 + Q3$).

Code availability: Statistical analysis was conducted using SAS[®] software version 9.4 (32), R Core Team (2023) version 4.3.1 (33) and Stata statistical software (release 18) (34). These codes can be provided by the corresponding author upon a reasonable request.

3. Results

During the first 10 years of follow-up, 778 out of 8501 children developed persistent confirmed IA. The median age (IQR) at risk was 36.3 (18.1, 72.5) months. Of the 778

children with IA, 590 had complete records. The incidence rate of Islet Autoimmunity was estimated to be 0.042 with 95% CI: (0.039, 0.045).

Descriptive statistics of the demographics are displayed by country, FDR status, HLA DR3/4 status and sex as shown in table 1 with the number of subjects at enrollment (age 3 months), number of person years follow-up, number of children developing IA with incidence, that were analyzed in the Cox model. The supplementary Table S1 displays additional descriptive statistics of the distribution of vitamin B_{12} intake including inter-quartile range (IQR), median, standard deviations and standard errors by country for several data sets used.

The trend of vitamin B_{12} levels (μg) by country and visit from 8,501 children in the TEDDY cohort followed regularly till 10 years of age while at risk of developing Islet Autoimmunity is shown in Figure 1, which shows that the intake/day of vitamin B_{12} varied on average between countries and by visit with the intake being consistently higher in the US and lowest in Germany. A box plot of the vitamin B_{12} intakes (Supplementary Figure S1) shows the presence of potential outliers at some of the follow-up visits.

Table 2 provides Cox regression estimates both in the log scale (log-hazard ratios) and in the exponential form (HR) together with the sandwich robust SE and the 95% CI. The HR represents the risk associated with a unit standard deviation (SD) change in take. The two outstanding extreme observations (vitamin B_{12} of 670.81 μg and 1666.83 μg) were scrutinized further by dropping each one of them in turn and re-analyzing the data, then later, dropping both values and repeated the analysis, and then replaced them with their medians calculated by country and visit and re-analyzed the data. Results indicated that dropping an observation from the full data with vitamin B_{12} intake of 670.81 $\mu\text{g/day}$ at the 6th year visit (visit 72 month) from USA had the largest impact on both the standard errors and the direction of the effect. It was noted that this observation was a confirmed Islet Autoimmunity positive case. The other outlier with vitamin B_{12} intake value of 1666.83 $\mu\text{g/day}$ was not an Islet Autoimmunity positive case and had minimal influence on the Cox regression model as shown in Table 2 under the sensitivity analysis sub-title.

After fitting the extended Cox regression models, residual diagnostics plots were examined for highly influential observations on the model parameters and are given in Figures 2, 3 and 4. Figure 2 displays eight residual diagnostics plots from 2(a) to 2(h) to find out any extreme observations that are influential. Figures 2(a) and 2(b) show standardized DFBETA residuals against vitamin B_{12} intake for full data and IQR-5 data. Extreme observations are noticeable from the full data. Similarly, Figures 2(c) and 2(d) display martingale residuals with potential outliers seen on the plot of full data. Deviance residual plots are provided in Figures 2(e) and 2(f) while score residuals are shown in Figures 2(g) and 2(h) for models fitted using the full data and IQR-5 data sets. All plots from the full data indicated the presence of at least two observations that have larger residuals than expected.

Figures 3 and 4 display partial DFBETA residuals by country for data analyzed based on the full data and IQR-5 data, respectively. We compared the magnitudes of the largest DFBETA values to the Cox regression coefficients and labelled the points by their vitamin B_{12} intake values. Results from Figure 3 showed that USA's vitamin B_{12} intake/day values of 670.81 μg

at visit 72 month and 1,666.83 μg at visit 120 month were disproportionately influential for the data from the full data but not from the IQR-5 dataset (Figure 4).

Similar patterns were observed when looking at the partial deviance residuals (Supplementary Figures S2 and S3), and partial score residuals plots (Supplementary Figures S5 and S6) by country. The goodness of fit graphs based on Cox-Snell residuals plotted against cumulative hazard for modeling the full data, IQR-5 data and $\log_2$ transformed data are provided in the Supplementary files (Figures S4(a), S4(b) and S4(c), respectively). There are light-heavy tails, but the models fitted the data reasonably well.

4. Discussion

The study has provided practical illustrations where IQR method can be used to identify potential outliers in longitudinal data sets by follow-up visit.

Analysis on TEDDY dietary data following the IQR method showed the impact of extreme observations on the model (Table 2). Results from the full data analysis indicated a significant increase in risk of developing IA with higher intakes of vitamin B_{12} . Analysis from the log transformed and IQR-k reduced datasets showed that an increase in intake of vitamin B_{12} reduces the risk of IA although the association between exposure and outcome in the Cox model was not statistically significant. Some of the detected outliers by the IQR method were also found to be highly influential observations by the residual diagnostic plots. The most influential extreme observation was found to be a case where the unusual large value of vitamin B_{12} of 670.81 μg was on the 6th year visit in USA when the subject was diagnosed with Islet Autoimmunity. Results in Table 2 show the impact of this outlier on the model. When this unusual observation is removed, the time-to-event analysis handles this subject as a non-case up to the last visit that appears in the dataset when data is arranged in the counting process format. As a result, the direction of the HR, the SE and the significance of the association changed drastically.

In many studies, researchers would not want to drop observations. We have illustrated the procedure of not having to drop outliers using these two extreme observations (vitamin B_{12} of 670 and 1666.83 μg) where we replaced them with their group-specific medians, and in using the log transformation and inverse weighting methods (Section 2.5). This made the 1 outlier out of the 590 cases remain as a case in the survival model with an intake value that was within the group range. Results for this analysis were seen to be in the same direction as for the IQR-k methods and $\log_2$ transformed method, showing a reduced risk of IA with intake of vitamin B_{12} and larger standard errors (0.216) than when the outlier are not handled (0.009).

The extreme value of vitamin B_{12} = 1666.83 μg was found to be on the 10th year visit in USA and although it is such an extremely larger value compared to the specific country and visit values, this observation had mild impact on the survival model since it was not a case. When only this outlier is removed, in the presence of the other case-outlier, the HR still indicated an increased risk of IA with intake of vitamin B_{12} .

Our study has shown that 1 outlier out of 590 cases of IA (Table 1) has a different impact on the model compared to 1 outlier out of 7911 (8501–590) non-cases. We could have had, say, 40 outliers out of 590 cases compared to 40 outliers out of 7911 non-cases. IQR-k algorithm can identify these potential outliers that could represent a genuine subgroup related to disease/case status. In survival regression models, cases with extreme values can be highly influential compared to non-cases.

Choices for the IQR scale factor k may depend on the distribution of the data and how far away from the median the researchers would like to keep the data points. Smaller k values are more stringent and make the distribution of the variable to be more precise than bigger k values which give room for larger variances. If no prior information on the distribution of the variable of interest is available, the analyst can examine several k values to identify potential outliers to flag them off then conduct sensitivity analyses to see if the results from the model parameters (estimate, standard errors, strength of association) change with or without the detected outliers.

Examining residual diagnostic plots for the full models compared to those from the IQR-5 models revealed similar patterns of the presence of potential outliers that could be highly influential in the full models but not in the IQR-5 models.

We have illustrated the use of IQR method to detect potential outliers of time-varying continuous variables in longitudinal data sets at follow-up visits. It can be used as a data quality control procedure to identify unusual observations. Once outliers are identified, they can be flagged off and investigated further, including conducting sensitivity analysis to ascertain their influence in the regression model.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data availability statement:

“Data from The Environmental Determinants of Diabetes in the Young (<https://doi.org/10.58020/y3jk-x087>) reported here will be made available for request at the NIDDK Central Repository (NIDDK-CR) website, Resources for Research (R4R), <https://repository.niddk.nih.gov/>.”

Abbreviations

Full data set	the original data set obtained after cleaning and applying quality check controls to fix obvious data error problems and unexpected observations.
IQR-k data set	where k=3,5,7, and 10 stands for data with extreme observations removed that were detected using scale factors k values.
IQR-k=3 weighted	The inversely weighted method described in Section 2.5 is used to replace detected outliers based on k=3 scale factor into values that are within the group range.
Log₂ data	is one that had the time dependent variable log transformed to base 2.
	Other full-datasets with one or two extreme observations removed and one other full-data set where the two extreme observations were replaced with the group specific medians calculated by country and visit are described in Table 2 under the sensitivity analysis section.

References

1. Agresti A, Franklin CA, Klingenberg B. Statistics: The Art and Science of Learning from Data. 5th ed. Pearson; 2021.
2. McClave JT, Sincich TT. Statistics. 13th ed. Pearson Higher Ed; 2017.
3. Aguinis H, Gottfredson RK, Joo H. Best-Practice Recommendations for Defining, Identifying, and Handling Outliers. *Organ Res Methods*. 2013 May;16(2):270–301.
4. Jones PR. A note on detecting statistical outliers in psychophysical data. Vol. 81, Attention, Perception, and Psychophysics. Springer New York LLC; 2019. p. 1189–96.
5. Leys C, Delacre M, Mora YL, Lakens D, Ley C. How to classify, detect, and manage univariate and multivariate outliers, with emphasis on pre-registration. *Int Rev Soc Psychol*. 2019;32(1).
6. Broeck J Van Den, Cunningham SA, Eeckels R, Herbst K. Data cleaning: Detecting, diagnosing, and editing data abnormalities. *PLoS Med*. 2005;2(10):966–70.
7. Stasinopoulos MD, Rigby RA, Heller GZ, Voudouris V, Bastiani F De. Flexible Regression and Smoothing Using GAMLSS in R. CRC Press; 2017.
8. Rigby RA, Stasinopoulos MD, Heller GZ, Bastiani F De. Distributions for Modeling Location, Scale, and Shape: Using GAMLSS in R. CRC Press; 2020.
9. Yang J, Rahardja S, Fränti P. Outlier detection: How to threshold outlier scores? In: ACM International Conference Proceeding Series. Association for Computing Machinery; 2019.
10. Van der Meer T, Te Grotenhuis M, Pelzer B. Influential cases in multilevel modeling: A methodological comment. *Am Sociol Rev*. 2010 Feb;75(1):173–8.

11. Yang S, Hutcheon JA. Identifying outliers and implausible values in growth trajectory data. *Ann Epidemiol.* 2016 Jan 1;26(1):77–80.e2. [PubMed: 26590476]
12. Leys C, Ley C, Klein O, Bernard P, Licata L. Detecting outliers: Do not use standard deviation around the mean, use absolute deviation around the median. *J Exp Soc Psychol.* 2013 Jul;49(4):764–6.
13. Phan HTT, Borca F, Cable D, Batchelor J, Davies JH, Ennis S. Automated data cleaning of paediatric anthropometric data from longitudinal electronic health records: protocol and application to a large patient cohort. *Sci Rep.* 2020 Dec 1;10(1).
14. Shi J, Korsiak J, Roth DE. New approach for the identification of implausible values and outliers in longitudinal childhood anthropometric data. *Ann Epidemiol.* 2018 Mar 1;28(3):204–211.e3. [PubMed: 29398298]
15. Dugravot A, Sabia S, Shipley MJ, Welch C, Kivimaki M, Singh-Manoux A. Detection of outliers due to participants' non-adherence to protocol in a longitudinal study of cognitive decline. *PLoS One.* 2015 May;10(7).
16. Boone-Heinonen J, Tillotson CJ, O'Malley JP, Marino M, Andrea SB, Brickman A, et al. Not so implausible: impact of longitudinal assessment of implausible anthropometric measures on obesity prevalence and weight change in children and adolescents. *Ann Epidemiol.* 2019 May;31:69–74.e5. [PubMed: 30799202]
17. Hazrati S, Hourigan SK, Waller A, Yui Y, Gilchrist N, Huddleston K, et al. Investigating the accuracy of parentally reported weights and lengths at 12 months of age as compared to measured weights and lengths in a longitudinal childhood genome study. *BMJ Open* [Internet]. 2016;6:11653. Available from: 10.1136/bmjopen-2016-011653
18. Farooqui T, Mustafa I, Christie T. Outliers in Educational achievement data: Their potential for the improvement of performance. *Pakistan J Stat.* 2014;30(1):71–82.
19. Voloh B, Watson MR, König S, Womelsdorf T. MAD saccade: Statistically robust saccade threshold estimation via the median absolute deviation. *J Eye Mov Res.* 2019;12(8).
20. Chen Z, Song S, Wei Z, Fang J, Long J. Approximating Median Absolute Deviation with Bounded Error. *Proc VLDB Endow* [Internet]. 2021 Jul;14(11):2114–2126. Available from: 10.14778/3476249.3476266
21. Casella G, Berger RL. *Statistical Inference.* 2nd ed. Duxbury; 2002.
22. Rousseeuw PJ, Croux C. Explicit Scale Estimators with High Breakdown Point. Dodge Y, editor. *\$L_1\$-Statistical Anal Relat Methods.* 1992;77–92.
23. TEDDY Study Group. The Environmental Determinants of Diabetes in the Young (TEDDY) Study. *Ann N Y Acad Sci* [Internet]. 2008;1150:1–13. Available from: 10.1196/annals.1447.062
24. Uusitalo U, Kronberg-Kippila C, Aronsson CA, Schakel S, Schoen S, Mattisson I, et al. Food composition database harmonization for between-country comparisons of nutrient data in the TEDDY Study. *J food Compos Anal.* 2011;24(4–5):494–505.
25. Cox DR. Regression models and life tables (with discussion). *J R Stat Soc B.* 1972;74:187–220.
26. Klein JP, Moeschberger ML. *Survival Analysis: Techniques for Censored and Truncated Data.* 2nd ed. Springer; 2003.
27. Hosmer DW, Lemeshow S, May S. *Applied Survival Analysis: Regression Modeling of Time-to-Event Data.* 2nd ed. John Wiley & Sons, Inc.; 2008.
28. Lin DY, Wei LJ. The robust inference for the cox proportional hazards model. *J Am Stat Assoc.* 1989;84(408):1074–8.
29. Zeger Scott L., Liang Kung-Yee. Longitudinal Data Analysis for Discrete and Continuous outcomes. *Biometrics.* 1986;42(1):121–30. [PubMed: 3719049]
30. Willett WC, Howe GR, Kushi LH. Adjustment for total energy intake in epidemiologic studies. *Am J Clin Nutr.* 1997;65(4 Suppl):1220S–1228S; discussion 1229S–1231S. [PubMed: 9094926]
31. Goldman AI, Carlin B, Crane LR, Launer C, Korvick JA, Deyton L, et al. Response of CD4 lymphocytes and clinical consequences of treatment using ddI or ddC in patients with advanced HIV infection. *J Acquir Immune Defic Syndr Hum Retrovirology.* 1996;11(2):161–9. [PubMed: 8556398]
32. SAS Institute Inc. *SAS Software 9.4 (SAS/STAT 15.2)* [Internet]. Cary, NC, USA; 2016. Available from: <http://www.sas.com/>

33. R Core Team. R: A Language and Environment for Statistical Computing [Internet]. Vienna, Austria; 2023. Available from: <https://www.r-project.org/>
34. StataCorp LLC. Stata Statistical Software. College Station, TX: StataCorp LLC; 2023.
35. Barnett V, Lewis T. Outliers in Statistical Data. 3rd ed. John Wiley & Sons, Inc.; 1994.
36. Cook RD. Detection of Influential Observations in Linear Regression. *Technometrics*. 1997;19(1):15–8.
37. Biggs J Assessment and Classroom Learning: a role for summative assessment? *Assess Educ Princ Policy & Pract*. 1998;5(1):103–10.

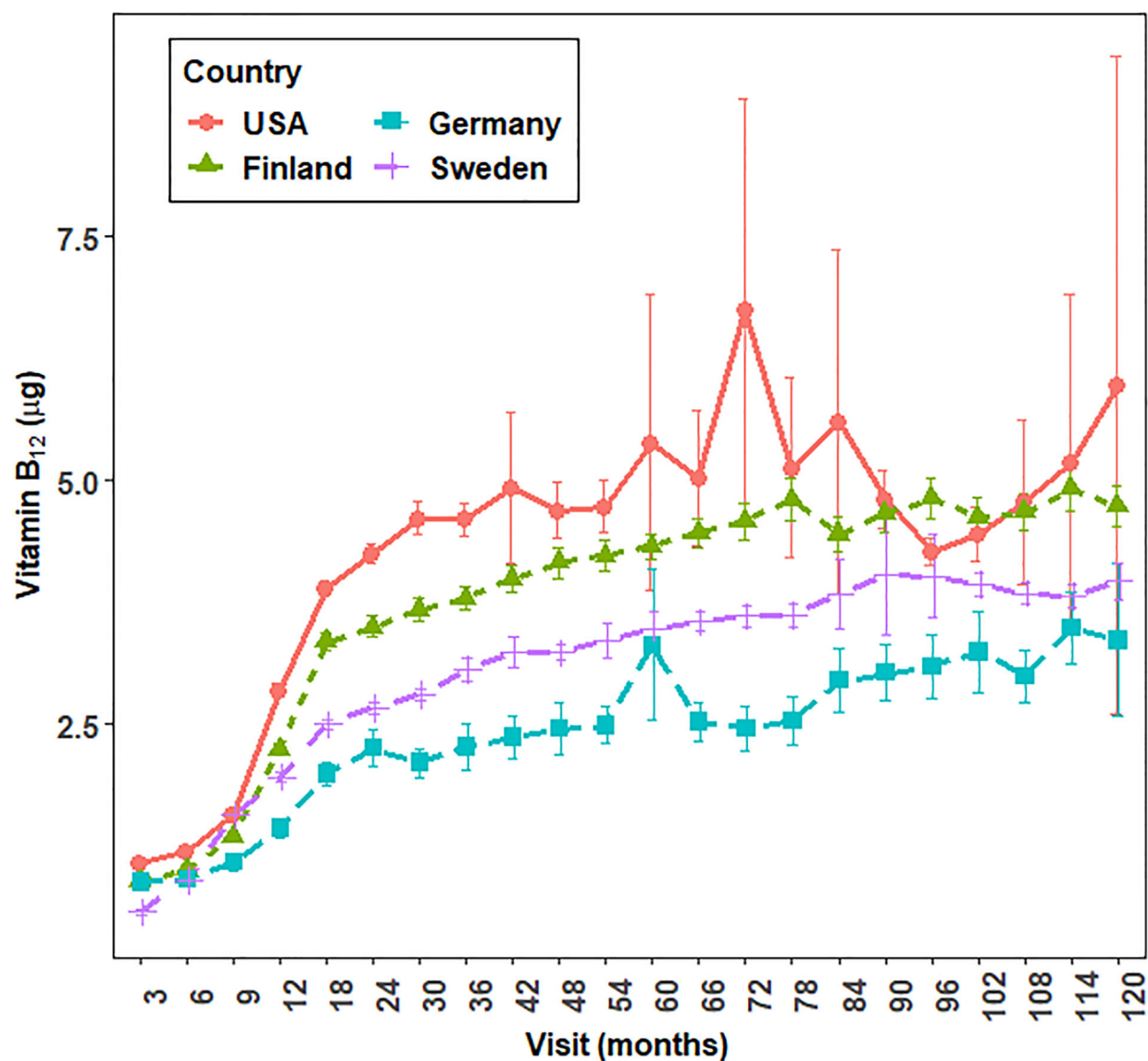


Figure 1:
Line graph of average vitamin B₁₂ (µg/day) intake by country in first 10 years of life.

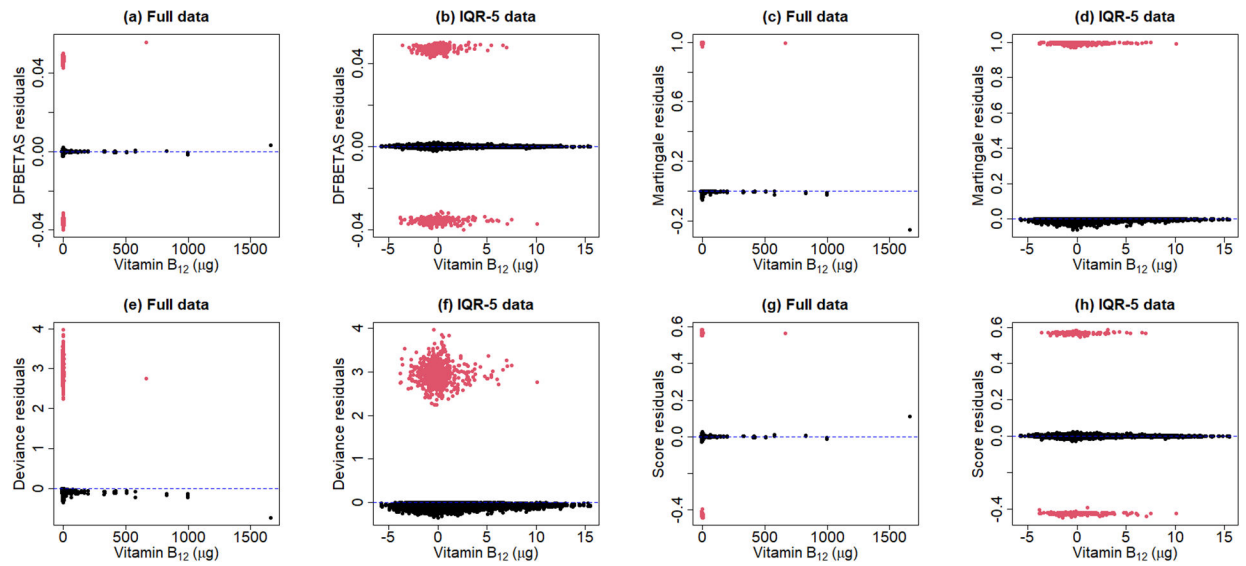


Figure 2:

Residuals diagnostics plots from fitted models for full and IQR-5 TEDDY data where residuals are plotted against average vitamin B_{12} ($\mu\text{g}/\text{day}$) intake/day. Figures 2(a) and 2(b) show the standardized DFBETA residuals, Figures 2(c) and 2(d) display the martingales residuals, Figures 2(e) and 2(f) are for deviance residuals and Figures 2(g) and 2(h) show score residuals against the vitamin B_{12} intake. Extreme observations that appear to be further away from other points are clearly seen on the full data plots.

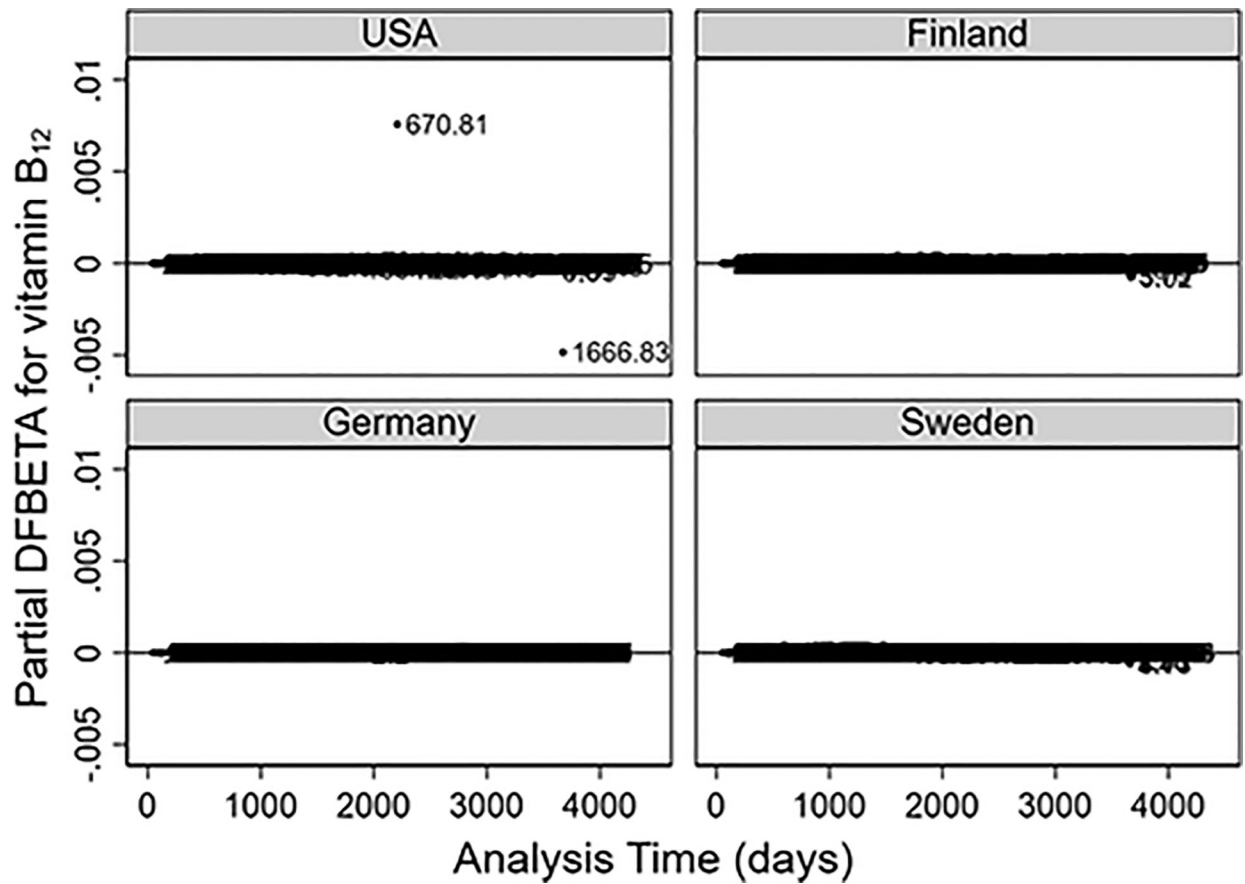


Figure 3.

Partial DFBETA residuals from full TEDDY dietary data showing vitamin B₁₂ ($\mu\text{g/day}$) values that have larger or unusual partial DFBETA residuals compared to other data points. The extreme observations from the full data come from USA in visits 72 month for the 670.81 μg and visit 120 months for the 1666.83 $\mu\text{g/day}$ intakes indicate to have a disproportionate influence on the fitted model.

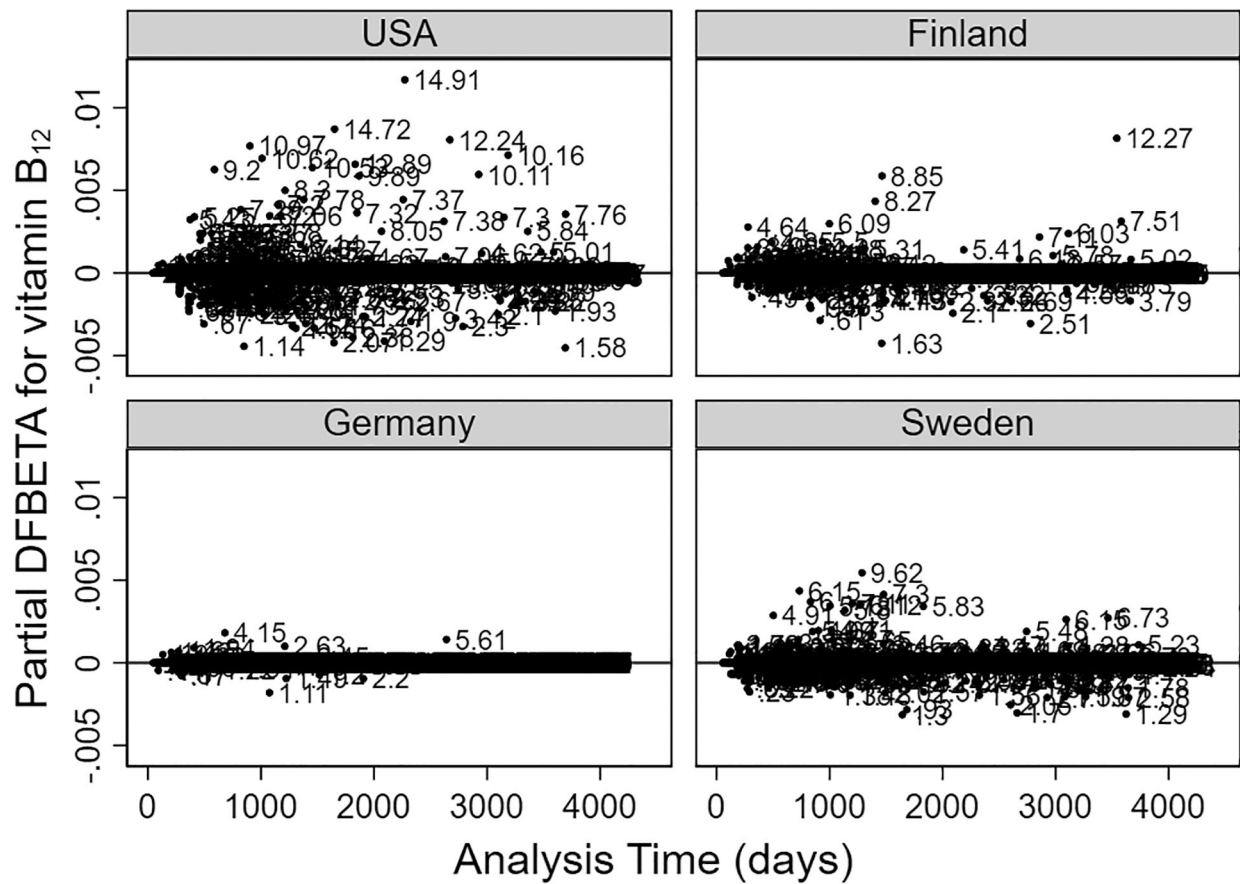


Figure 4.

Partial DFBETA residuals from IQR-5 TEDDY data with labeled vitamin B_{12} ($\mu\text{g/day}$) values.

There are no obvious extreme values shown on the residual plot.

Table 1.
Demographics of study participants and the risk of getting Islet Autoimmunity where subjects are followed-up for the first 10 years of life.

Variable	Total subjects followed from enrollment at age 3 months (N=8501)	Developed IA by age 10 years (N=590)
Incidence rate (per 1000 person/years); 95%CI		0.042 (0.039, 0.045)
IA age at risk month; median (Q1, Q3)		36.3 (18.1, 72.5)
Country		
USA	3616 (42.5 %)	207 (35.1 %)
Finland	1803 (21.2 %)	139 (23.5 %)
Germany	580 (6.8 %)	30 (5.1 %)
Sweden	2502 (29.4 %)	214 (36.3 %)
Family history with T1D		
no	7540 (88.7 %)	477 (80.9 %)
yes	961 (11.3 %)	113 (19.1 %)
Human Leukocyte Antigen (HLA) genotype DR3/4		
no	5178 (60.9 %)	299 (50.7 %)
yes	3323 (39.1 %)	291 (49.3 %)
Sex		
Male	4305 (50.6 %)	336 (56.9 %)
Female	4196 (49.4 %)	254 (43.1 %)

Table 2.

Association of vitamin B_{12} ($\mu\text{g/day}$) Intake/day with risk of Islet Autoimmunity (IA) from TEDDY dietary data.

Data set (obs.)	Vitamin B_{12} obs. dropped n (%)	β (SE)	HR (95% CI)	P-value
Full (n=152,426)	0 (0%)	0.024(0.009)	1.025 (1.006,1.044)	0.010
IQR-k=10 (n=152,426)	77 (0.05%)	-0.030(0.044)	0.971 (0.891,1.057)	0.496
IQR-k=7 (n=152,426)	134 (0.09%)	-0.023(0.044)	0.977 (0.897,1.065)	0.596
IQRk=5 (n=152,400)	243 (0.16%)	-0.013(0.044)	0.987 (0.905,1.077)	0.773
IQR-k=3 (n=152,217)	811 (0.53%)	-0.006(0.045)	0.994 (0.910,1.086)	0.897
IQR-k=3 wtd (n=152,217)	0 (0%)	-0.004(0.045)	0.996 (0.912,1.088)	0.930
Log2 transformed (n=152,426)	0 (0%)	-0.019(0.054)	0.982 (0.884,1.09)	0.728
Sensitivity analysis after residual diagnostics				
[†] Full data -case extreme obs. (n= 152,425)	1 (0 %)	-0.240(0.211)	0.787 (0.52,1.189)	0.255
[‡] Full-non-case with extreme obs. (n= 152,425)	1 (0 %)	0.035(0.012)	1.035 (1.011,1.06)	0.004
[*] Full less two extremes Obs. (n= 152,424)	2 (0 %)	-0.215(0.189)	0.807 (0.557,1.168)	0.255
[*] Full data extremes replaced with medians (n=152,426)	0 (0 %)	-0.252(0.215)	0.777 (0.509,1.185)	0.242

Hazard ratios represent the risk associated with a unit standard deviation (SD) change in intake. The Estimate (β) is the log-hazard ratio shown with the robust standard errors (SE). HR are hazard ratios provided with 95% confidence intervals. Several data sets were used under different IQR-K conditions. For sensitivity analysis:

[†] Vitamin B_{12} =670.81 μg from visit 72 month in USA was dropped from the full data.

[‡] Vitamin B_{12} =1666.83 μg value from visit 120 month from USA was dropped from the full data.

^{*} Both values of vitamin B_{12} were dropped (670.81 μg and 1666.83 μg).

^{*} The two extreme values were replaced with their specific by-country and visit median values.